

TECHNICAL PROJECT REPORT

Flight Controller Requirement Standardization System

Deterministic Requirement Analysis, Compliance Gating and Controlled AI-Assisted Candidate Generation

Project Item	Current Description
Project Type	Requirements Engineering System
Core Processing	Deterministic Rule Checking + Controlled AI-Assisted Candidate Generation
Deployment Context	Restricted DRDO Environment with no external egress except the configured trusted vLLM endpoint
AI Integration	DRDO-internal vLLM server using an OpenAI-compatible HTTP API
Project Scope	Flight Controller Requirement Workbooks
Prepared for	Technical Review

1. Introduction

Flight controller development depends on requirements that are precise, testable, unambiguous and consistently written. In practice, requirement workbooks often contain statements written by different authors over long development cycles. This creates variation in wording, structure and quality. A requirement may express the correct engineering intent while still containing vague language, multiple thoughts, open-ended clauses, missing measurable criteria or inconsistent terminology.

1.1 Problem Statement

Manual review of large requirement workbooks is time-consuming and difficult to scale consistently. The project therefore needs a system that can analyse requirements deterministically, identify objective defects quickly, avoid unnecessary AI calls, and present reviewable outputs to an engineer. The updated deployment model must operate within a restricted environment and communicate only with a trusted, explicitly configured DRDO-internal vLLM endpoint when AI-assisted operations are requested.

1.2 Existing Solutions and Gap

- Manual requirement review provides engineering judgement but becomes slow and inconsistent for hundreds of statements.
- Rule-only validation is fast and transparent, but cannot always produce a well-phrased alternative requirement.
- Generation-first approaches can improve wording, but running AI generation for every requirement wastes compute and networked inference capacity.
- A fully local-model design adds runtime and hardware management overhead on the deployment machine.

1.3 Solution Explored

The solution combines deterministic validation with controlled AI-assisted candidate generation. The deterministic path performs inexpensive structural and quality screening first. Only actions that require AI assistance contact the configured DRDO-internal vLLM service through the OpenAI-compatible chat completions API. The rest of the workflow remains available without an LLM connection, and final review remains under human control.

Objective	How the System Addresses It
Restricted deployment	No external cloud dependency; AI calls are limited to the one trusted configured vLLM endpoint.
Fast early screening	Deterministic rules run independently of the LLM service.
Controlled AI usage	Generation and selected AI-assisted checks are invoked only when a feature explicitly requires them.
Graceful degradation	Core analysis, manual editing and Excel export

	continue when the AI endpoint is unavailable.
Traceability	Scores, rule findings, rejection reasons, progress and timings are recorded.
Human oversight	Outputs remain available for engineering review rather than being silently accepted.

2. Methodology

2.1 Solution Approach

The methodology separates processing into a deterministic validation path and an optional AI-assisted path. Deterministic checks are fast, explainable and independent of network reachability. AI-assisted operations are sent to the configured DRDO-internal vLLM server only when a feature explicitly requires them.

This separation ensures that loss of the AI service does not stop the complete application. A user can upload a workbook, perform EARS and deterministic INCOSE analysis, inspect violations, manually edit a requirement and export the reviewed workbook without any active LLM connection.

2.2 Processing Pipeline

Stage	Purpose	Execution Type
1. Parse	Extract and normalize requirement text from the workbook.	Deterministic
2. RuleFlag	Detect immediately identifiable requirement defects.	Deterministic
3. ClassifyPattern	Classify the requirement against supported structural patterns.	Deterministic
4. AbbreviationCheck	Check acronym and abbreviation issues covered by R37/R38.	Deterministic
5. IncoseCheck	Apply implemented deterministic INCOSE quality rules and report violations.	Deterministic
6. ComplianceCheck	Validate EARS-related structural compliance.	Deterministic
7. AI-Assisted Actions	Generate rewrites, run accurate scoring where requested, and perform contradiction checks.	vLLM HTTP Service
8. Score / Recommend	Score available results and recommend the best outcome where applicable.	Deterministic + optional AI-assisted results
9. Review / Finalize	Expose result for review,	Human-in-the-loop

	manual editing and export.	
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2.3 Assumptions

- The input workbook contains identifiable requirement statements that can be parsed into individual records.
- The deterministic path can operate without any LLM server.
- AI-assisted features require network reachability to the configured trusted DRDO vLLM endpoint.
- A human reviewer remains responsible for accepting or rejecting the final engineering requirement.

3. Requirements Checker Suite

3.1 Deterministic Checking

Deterministic checks are the first and continuously available line of processing. They do not require the vLLM endpoint and therefore provide low latency and explainable outcomes. Each detected issue can be associated with a rule, condition or structural reason.

- RuleFlag detects configured requirement defects.
- ClassifyPattern determines the relevant EARS-style pattern classification.
- AbbreviationCheck handles acronym and abbreviation checks corresponding to R37 and R38.
- IncoseCheck evaluates the implemented deterministic portion of the INCOSE rule set and returns score and violations.
- ComplianceCheck validates the structural EARS gate.

3.2 AI-Assisted Operations

The AI client now communicates with a DRDO-internal vLLM server rather than running Ollama locally. Generation uses POST {base_url}/chat/completions and reachability is checked using GET {base_url}/models. The configured service exposes an OpenAI-compatible chat completions interface.

The AI-assisted operations include requirement rewrite generation, the Check Accurate Score action for the rules that require model judgement, and the consistency page's contradiction check. These operations are optional from the perspective of the core deterministic workflow.

3.3 Why the Split is Important

Characteristic	Deterministic Path	AI-Assisted vLLM Path
Availability	Works with zero LLM connection.	Requires configured endpoint to be reachable.
Explainability	Direct rule IDs, scores and specific failure reasons.	AI response is used for the requested generation or

		judgement task.
Compute / infrastructure	Runs locally with low processing overhead.	Uses shared DRDO-internal vLLM inference infrastructure over HTTP.
Best use	Filtering, checking, scoring, manual review support and export.	Rewrite generation, accurate scoring and contradiction checking.

4. Pre-Generation INCOSE Compliance Gate

4.1 Role in the Updated Architecture

The INCOSE compliance gate remains part of the deterministic quality-control architecture. It evaluates the original requirement using the implemented automatable rules before unnecessary AI-assisted processing. The purpose is to preserve fast and explainable quality screening while reserving AI service usage for operations where model judgement or generation adds value.

4.2 Pipeline Order

Core Flow	Updated AI-Enabled Flow
Parse → RuleFlag → ClassifyPattern → AbbreviationCheck → IncoseCheck → ComplianceCheck	Core deterministic flow remains available without AI connectivity.
After deterministic analysis	Generate / Check Accurate Score / Contradiction Check use the configured vLLM service only when requested.

4.3 Gate Behaviour

The deterministic INCOSE evaluation preserves detailed rule information rather than returning only a numeric score. Findings can identify issues such as vague terms, open-ended clauses, multiple 'shall' statements, combinators, absolutes, universal qualification problems or missing measurable criteria. The EARS structural check remains separately available to validate the expected requirement pattern.

5. AI Service Migration and Configuration

5.1 Ollama to vLLM Migration

The project dropped the local Ollama runtime entirely and now communicates with a DRDO-internal vLLM server over plain HTTP using the OpenAI-compatible chat completions API. No new Python dependency was required because httpx was already used by the project. The internal client interface was preserved so existing callers continue to use the same operational methods.

Area	Current Implementation
Generation	POST {base_url}/chat/completions
Reachability check	GET {base_url}/models
Client class	LocalLLMClient name retained for caller compatibility
Public methods	check_reachable, generate_structured and generate_json remain unchanged
Error rename	OllamaUnavailableError mechanically renamed to LLMUnavailableError
New dependency	None; existing httpx transport is used
Authentication	Empty api_key means no Authorization header is sent

5.2 Current Configuration

Setting	Value	Note
base_url	http://80.1.0.24:8001/v1	Corrected internal IPv4 endpoint.
model	gemma4-26B-A4B-it	Must exactly match the name reported by GET {base_url}/models.
api_key	"" (empty)	No Authorization header is sent.
request_timeout_seconds	900	Allows for slower responses from a shared/networked server.
temperature	0.2	Controls generation variability.

5.3 Deployment Model

The deployment model is no longer described as fully air-gapped or loopback-only. The correct operational model is restricted execution with no external egress except communication to the one trusted and configured DRDO vLLM endpoint. Network guard logic, Sidebar/Settings/Dashboard copy and error messages were updated accordingly.

6. Test Execution and Result Scenarios

6.1 Core Deterministic Scenario

With no LLM server available, the application can still upload a workbook, classify EARS patterns, perform deterministic INCOSE analysis and report violations, allow manual requirement edits, and download the Excel export. None of these core paths requires config/settings.yaml's llm.base_url.

Action	Behaviour Without AI Service
Workbook upload	Works normally.
EARS pattern classification	Works normally.
Implemented 28-rule INCOSE score and violations	Works normally.
Manual requirement editing	Works normally.
Excel export	Works normally.
Generate rewrite	Fails cleanly with AI_SERVICE_UNAVAILABLE if endpoint is unreachable.
Check Accurate Score	Fails cleanly only for the requested action if endpoint is unreachable.
Contradiction check	Fails cleanly only for the requested action if endpoint is unreachable.

6.2 AI Service Available Scenario

When the configured endpoint is reachable and the model name is valid, the Generate action sends OpenAI-compatible chat completion requests to the vLLM server. The accurate scoring and consistency contradiction-check workflows can similarly obtain model-assisted results.

6.3 AI Service Unavailable Scenario

If the endpoint is unreachable, the requested AI-assisted action returns a clear AI service unavailable condition. The failure is isolated to that action or row; the rest of the deterministic application continues to function. This preserves graceful degradation and avoids turning an inference-service outage into a complete application outage.

7. Results and Performance Observations

7.1 Deterministic Path Performance

The deterministic rule engine remains independent of the AI transport. Rule checks, EARS-related validation, deterministic INCOSE scoring and violation reporting are not delayed by vLLM availability. This makes the deterministic path suitable for fast analysis and transparent review.

7.2 Networked AI-Assisted Performance

AI-assisted response time is now influenced by shared vLLM server availability, network latency, queueing, model execution and prompt complexity rather than by a local Ollama process on the deployment machine. The configured 900-second timeout reflects the possibility of slower responses from shared inference infrastructure.

7.3 Result of the Migration

Processing Area	Observed / Intended Behaviour
Deterministic analysis	Independent of AI service reachability.
AI generation	Uses the configured DRDO vLLM OpenAI-compatible endpoint.
API compatibility	Existing LocalLLMClient callers require no interface changes.
Failure isolation	AI unavailability affects only AI-assisted actions.
Deployment dependency	No local Ollama runtime is required.
Offline transfer bundle	Ollama package dependency was removed from requirements.txt.

8. Inference and Discussion

8.1 Main Inference

The migration separates application availability from local model runtime management. Core requirement engineering functions remain deterministic and available without AI connectivity, while model-dependent capabilities are explicitly isolated behind the trusted internal vLLM service.

8.2 Why Graceful Degradation Matters

A restricted engineering environment may temporarily lose access to the inference endpoint without losing the ability to review a workbook. By isolating generation and model-judgement features, engineers can continue analysis, editing and export work instead of waiting for the AI service to recover.

8.3 Practical Impact for Deployment

Removing the local Ollama runtime reduces deployment-machine requirements and removes one Python package from the transfer bundle. The trade-off is that AI-assisted features now depend on reachability to the configured internal endpoint and on the server exposing the expected model name.

8.4 Explainability and Review Value

Deterministic violations remain directly traceable to implemented rules and reasons. AI-assisted results are used as additional workflow capabilities rather than replacing the deterministic evidence available to the reviewer.

9. Limitations and Constraints

9.1 Internal Network Dependency

The application is not fully air-gapped for AI-assisted functions. Generate, accurate scoring and contradiction checking require reachability to the configured DRDO vLLM endpoint. Core deterministic functions remain available when that dependency is absent.

9.2 Endpoint and Model Configuration

Two deployment checks remain important in the target environment: the configured base URL must be reachable, and the configured model string must exactly match the value reported by GET {base_url}/models. These checks should be completed before relying on AI-assisted features during deployment.

9.3 Rule Coverage Constraint

Only rules suitable for deterministic automation are fully enforced by the local rule engine. Context-dependent aspects of requirement quality continue to require engineering judgement or the optional AI-assisted workflows.

9.4 Shared Service Variability

Because inference is provided by a shared networked server, response time can vary according to server load, model scheduling, network conditions and request complexity. The application therefore uses explicit timeouts and clear service-unavailable handling.

10. Future Work

10.1 Verification in the Target Environment

- Run GET `http://80.1.0.24:8001/v1/models` from the target environment and confirm endpoint reachability.
- Verify that `gemma4-26B-A4B-it` exactly matches a model identifier returned by the server.
- Perform end-to-end validation of Generate, Check Accurate Score and contradiction checking against the target vLLM service.
- Record latency and availability observations under realistic shared-server usage.

10.2 Workflow Improvements

- Provide clearer UI status indicators showing whether deterministic-only operation or AI-assisted operation is currently available.
- Add batch-level summaries for deterministic pass/fail outcomes and AI-assisted actions attempted.
- Expand export reporting with requirement IDs, rule findings, EARS outcomes, AI-assisted results and reviewer decisions.
- Continue isolating optional AI operations so service interruptions do not affect the core review workflow.

10.3 Conclusion

The Flight Controller Requirement Standardization System now combines deterministic requirement analysis with controlled access to a DRDO-internal vLLM inference service. The migration removes the local Ollama runtime while preserving the existing client interface and graceful failure behaviour. Core functions such as workbook upload, EARS classification, deterministic INCOSE scoring and violations, manual editing and Excel export remain operational without any LLM connection. Only explicitly AI-assisted actions depend on the configured endpoint. The updated architecture therefore supports restricted deployment while maintaining clear separation between deterministic engineering controls and optional model-assisted capabilities.